

Snare Drum Synthesizer

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The method described below may differ from the frequently updated GitHub repository implementing it.

Abstract

This document presents a custom, simple and intuitive drum snare synthesizer designed without the need for an audio synthesis background. Starting from a single real acoustic sample, the objective is to imitate the sound through a step-by-step approach.

1 Sample Analysis

The analysis of the target sample relies solely on its waveform. It reveals clear characteristics: a sine wave that loses amplitude over time, accompanied by noise. This noise is distinctly concentrated at the peaks and troughs of the fundamental sine wave.

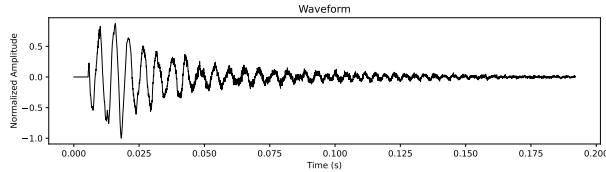


Figure 1: Target sound waveform

The interpretation is straightforward: the main waveform corresponds to the shell sound, the drum's core resonance, while the noise comes from the snare wires striking the membrane periodically from below. Hence the two-component model described below.

2 Model

2.1 The Shell (Drum Body)

The fundamental tone of the drum is modeled as a sine wave of frequency f , modulated by an envelope function E , centered on a slight impact instant i :

$$y(t) = \sin(2\pi ft) \cdot E(t - i) \quad (1)$$

The choice of E is driven mainly by how fast the amplitude decays right after the impact, which naturally points to the exponential family: a single scale parameter directly controls this rate. Two immediate candidates are the Gaussian envelope — the more intuitive choice at first glance,

$$e^{-\frac{t^2}{2b^2}}, \quad (2)$$

and the Laplace (double-exponential) envelope,

$$e^{-\frac{|t|}{b}}. \quad (3)$$

The Laplace envelope was retained empirically: overlaid on the noise-free target waveform, its sharper decay around the impact reproduces the observed decay far more closely than the smoother Gaussian roll-off (Figure 2).

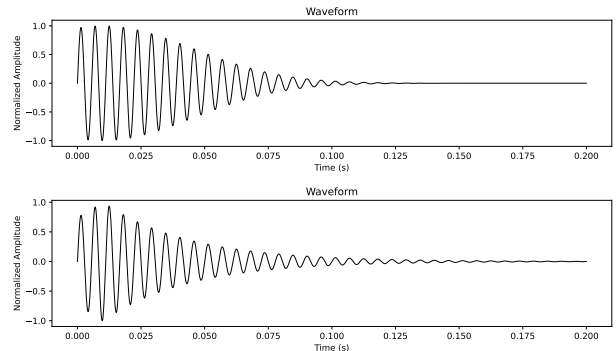


Figure 2: Noise-free shell waveform overlaid with the fitted Gaussian (top) and Laplace (bottom) envelopes.

The shell model is therefore

$$y(t) = \sin(2\pi ft) \cdot e^{-\frac{|t-i|}{b}} \quad (4)$$

Adjusting b directly controls the "dryness" of the hit. It is worth noting that, to the human ear, this simple model almost perfectly reproduces the sound of a snare drum *without* its snare wires, which provides a strong baseline to build on.

2.2 The Noise (Snare Wires)

To model the snare wires, an elementary noise block is required. Reusing the Laplace envelope introduced above (the Gaussian candidate is discarded), the block is defined as a uniform distribution modulated by a Laplace envelope:

$$N(t) = U(-1, 1) \cdot e^{-\frac{|t-i|}{b_N}} \quad (5)$$

where $U(-1, 1)$ is a uniform random variable and b_N is the noise's own Laplace scale. Both the centering instant i (taken at each peak of the shell's oscillation) and b_N are formalized in the next section.

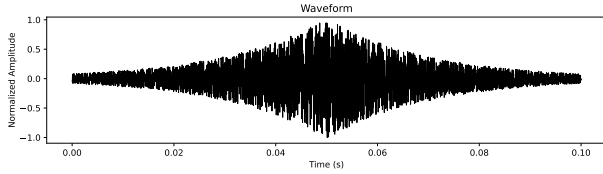


Figure 3: Waveform of an isolated noise block.

2.3 Synthesizer Logic

The full signal combines the shell with noise blocks inserted at the peaks $t_2, \dots, t_n$ of the shell's oscillation (t_1 is kept aside, used only to anchor the sampling probability below). Since each block's own envelope already concentrates it tightly around its peak, every term is effectively an indicator localized on a window of half-width T around that peak. With a constant amplitude A_0 as a starting point, the signal is formalized as

$$s(t) = y(t) + \sum_{k=2}^n A_0 \cdot \mathbb{1}_{[t_k-T, t_k+T]}(t) \cdot U_k(-1, 1) \cdot e^{-\frac{|t-t_k|}{b_N}} \quad (6)$$

The final step is to align these bursts with the shell's peaks. Anchoring them at random instants, even at a fixed average rate, produces a detached, unnatural sound (empirically observed). Bursts are instead anchored directly on the peaks; this choice makes no audible difference toward the end of the signal, where the sampling probability below is already close to one and noise is present around virtually every peak regardless of exact alignment.

Visual inspection of the acoustic waveform shows that noise appears less frequently around the initial peaks, with its frequency of occurrence increasing toward the tail of the sound (approaching one occurrence per peak). This behavior is modeled through

Bernoulli sampling, with an increasing probability of noise generation across successive peaks. This probability grows exponentially, starting at p_1 at the first peak t_1 and reaching 1 at a reference peak t_c . A linear growth was found to be too slow, so an exponential rate was chosen instead, leading to the following system for $A, \beta \in \mathbb{R}$:

$$A \cdot e^{\beta t_1} = p_1 \quad (7)$$

$$A \cdot e^{\beta t_c} = 1 \quad (8)$$

Solving this system in closed form gives

$$\beta = \frac{-\ln p_1}{t_c - t_1}, \quad A = p_1 e^{-\beta t_1} \quad (9)$$

Finally, the noise's scale b_N is chosen so that a fraction α of the envelope's area is concentrated within the window $[t_k - T, t_k + T]$, where $T = \frac{1}{4f}$ is a quarter of the shell's period (i.e., half of the noise block's total duration, itself set to half the shell's period). Since $|t - t_k|$ follows a Laplace distribution of scale b_N , the probability mass contained in this window is

$$P(|t - t_k| < T) = 1 - e^{-\frac{T}{b_N}} \quad (10)$$

Setting this fraction to α and solving for b_N gives

$$1 - e^{-\frac{T}{b_N}} = \alpha \implies b_N = \frac{T}{\ln\left(\frac{1}{1-\alpha}\right)} = \frac{1}{4f \ln\left(\frac{1}{1-\alpha}\right)} \quad (11)$$

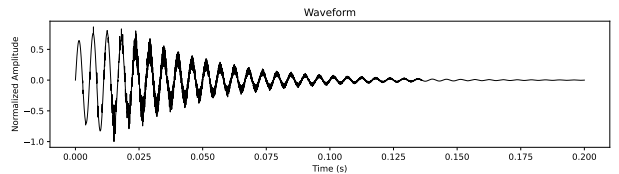


Figure 4: Full synthesized snare waveform.

Every equation above was expressed in terms of literal parameters. Two categories of values complete the model:

- **Varying across generation:** the frequency f and the shell's Laplace scale b .
- **Fixed:** the impact instant $i = 0.01$, the constant noise amplitude $A_0 = 0.5$, the initial sampling probability $p_1 = 0.01$, the reference peak index $c = 15$, and the area fraction $\alpha = 0.5$.

3 Sample Generation

Sweeping the frequency f from 150 to 250 Hz in steps of 10 Hz, and the decay $\frac{1}{b}$ from 10 to 50 in steps of 4, yields 500 distinct synthesized samples. Figures 5, 6, and 7 show three examples, synthesized respectively with $(f, \frac{1}{b}) = (180, 30)$, $(180, 10)$, and $(250, 30)$.

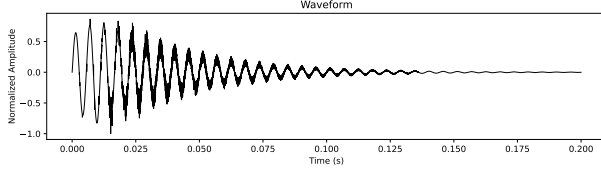


Figure 5: $(f, \frac{1}{b}) = (180, 30)$

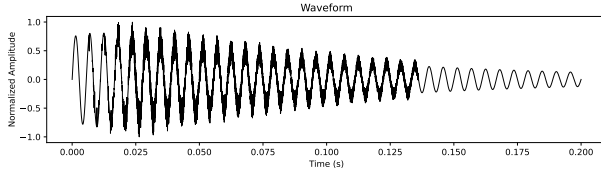


Figure 6: $(f, \frac{1}{b}) = (180, 10)$

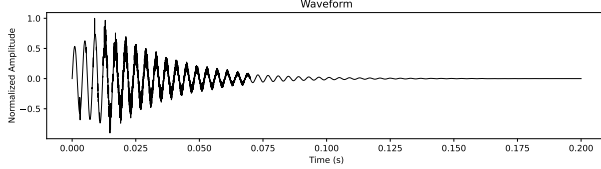


Figure 7: $(f, \frac{1}{b}) = (250, 40)$

4 Synthesis Review & Future Work

While every frequency choice is legitimate — it mostly affects the perceived pitch of the sound — decays close to 10 are not convincing: the noise becomes over-represented and the overall sound feels artificial. Figure 6 (decay = 10) illustrates this: even the sound’s duration ends up too long. However, for decays roughly between 25 and 40, the resulting samples are convincing synthetic snare drums, with noticeable variety across the different settings. The result is still far from a fully realistic acoustic sound, but the simplicity of the model is satisfying enough to produce a usable raw snare sound. It can readily be mastered, EQ’d, or compressed, and incorporated into mixes for pop, alternative, or rap productions.

The quality of the synthesis can be assessed, for instance, by listening to the sample with decay 30 and frequency 180, and by comparing its waveform and spectrogram to the target’s in Figure 9.

For future work: rather than trying to further improve the approximation — which is hard to do from waveform analysis alone — one could use this basic synthesizer in reverse, to detect snares in electronic songs. This would involve training a detection model on the noisy samples. For example, the detection model could be a CNN applied to the spectrogram of the noisy samples, later tested on real songs to check whether it manages to detect snares. A good quality metric for such a model would be near-perfect precision, even at the cost of low recall. A model with close to 100% precision could isolate a small but reliable set of true snare hits from a wide variety of songs; these samples could in turn be used to train a more accurate generative model.

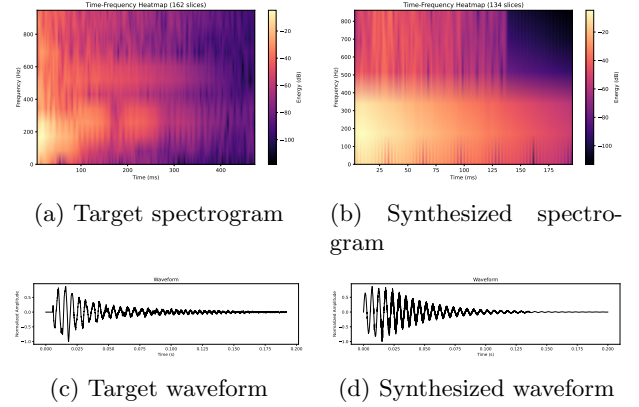


Figure 9: Target vs. synthesized snare drum: spectrograms and waveforms.